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(The proof of equation (8.39) left as an exercise).

EXAMPLE 8.18



We often have the experience of pumping air into bicycle tyre using hand pump. Consider the air inside the pump as a thermodynamic system having volume V at atmospheric pressure and room temperature, 27°C. Assume that the nozzle of the tyre is blocked and you push the pump to a volume 1/4 of V. Calculate the final temperature of air in the pump? (For air , since the nozzle is blocked air will not flow into tyre and it can be treated as an adiabatic compression).

Solution

Here, the process is adiabatic compression. The volume is given and temperature is to be found. we can use the equation (8.38)

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}.$$

$$T_i = 300\ K \quad (273+27°C = 300\ K)$$

$$V_i = V \ \&\ V_f = \frac{V}{4}$$

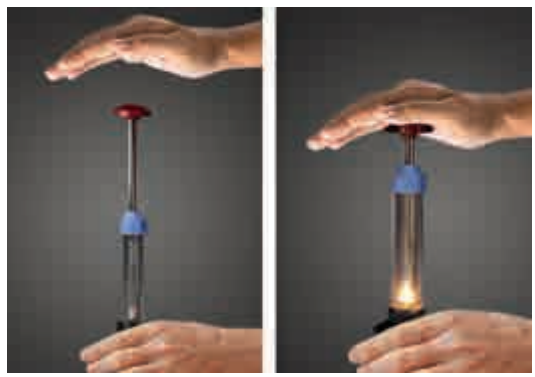
$$T_f = T_i \left(\frac{V_i}{V_f}\right)^{\gamma-1} = 300\ K \times 4^{1.4-1} = 300K \times 1.741$$

T2 ≈ 522 *K* or 249^{0}C

This temperature is higher than the boiling point of water. So it is very dangerous to touch the nozzle of blocked pump when you pump air.



When the piston is compressed so quickly that there is no time to exchange heat to the surrounding, the temperature of the gas increases rapidly. This is shown in the figure. This principle is used in the diesel engine. The air-gasoline mixer is compressed so quickly (adiabatic compression) that the temperature increases enormously, which is enough to produce a spark.



Work done in an adiabatic process:

Consider μ moles of an ideal gas enclosed in a cylinder having perfectly non conducting walls and base. A frictionless and insulating piston of cross sectional area A is fitted in the cylinder as shown in Figure 8.31.

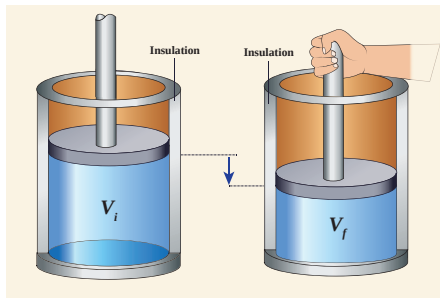


Figure 8.31 Work done in an adiabatic process

Let W be the work done when the system goes from the initial state (P_i, V_i, T_i) to the final state (P_f, V_f, T_f) adiabatically.